PRODUCT REQUIREMENTS DOCUMENT

Document Version	1.0
Product Name	Blur Or Bokeh
Document Owner	Eric Julianto
Developer	Amalan Fadil, Nafis, Tegar
Stakeholder	
Doc Stage	Ongoing
Created Date	2024-02-19

Overview

Problem Statement
In e-commerce and online marketplace platforms, ensuring the quality of product/service display photos is essential for maintaining user trust and enhancing the overall shopping experience. However, distinguishing between clear, high-definition (HD) images and blurred photos, which may resemble intentional bokeh effects, poses a challenge for content moderation. Without an efficient system to automatically identify and flag blurred images, platforms risk displaying subpar content to users, leading to dissatisfaction and potential loss of credibility. There is a critical need for a Blur or Bokeh detection service that accurately assesses the quality of uploaded product/service display photos, automatically banning blurred images to uphold content quality standards and optimize user experience.

Objective
This project aims to develop a Blur or Bokeh detection service for product/service display photos uploaded to e-commerce and online marketplace platforms. Leveraging image processing, machine learning algorithms, and computer vision, the service will analyze uploaded photos to determine whether they exhibit characteristics of clear HD images or blur, with consideration for potential bokeh effects. Photos identified as blurred will be automatically flagged and banned from display on the platform, ensuring that only high-quality images are presented to users. By implementing this service, platforms can enhance content moderation efforts, improve user satisfaction, and maintain a reputable online marketplace environment.

CRISP DM
In this project, we will implement the framework that already been implemented in previous project, Cartoon Or Not

DARCI

	Tags	Guidelines

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Decider	Eric Julianto	Make a features and approver major decisions related to the development and deployment of the Blur Or Bokeh project
Accountable	Alfizdiana Sholikhah	Overseeing and managing the completion of tasks assigned to team members
Responsible	Amalan Fadil, Eric Julianto, Fauzan Ihza Fajar, Nafis, Tegar	Actively involved in tasks related to feature development.
Consulted	Eric Julianto	It is essential to feel at ease and in agreement with the progress of feature development, while also seeking input and feedback. The decision-makers and those responsible must guarantee the active involvement of those being consulted.
Informed	Eric Julianto	It is important to be comfortable and aligned with the progress of feature development, while actively seeking input and feedback. Decision-makers and those in charge should ensure the involvement of those consulted.

Project Timeline

Time Period	Activity	PIC
19 Feb - 24 Feb	Gathering Dataset	Nafis, Tegar, Fadil
26 Feb - 1 Mar	Modelling	Nafis, Tegar, Fadil
4 Mar - 8 Mar	Creating Package	Nafis, Tegar, Fadil
11 Mar - 15 Mar	Deployment	Nafis, Tegar, Fadil

Success Matrics

Metrics	Definition	Actual	Target
Model Accuracy	The percentage of images correctly classified by the model as either cartoons or real-life photographs.		95%
Processing Speed and API Response Time	The time taken for the model to classify a single image.		< 4 seconds
User Satisfaction	Overall satisfaction level of users with the tool's performance, ease of use, and support.		4/5

User Stories

Title	User Story	Acceptance Criteria	Priority

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Dataset Access	As a user, I want to access a comprehensive dataset of images for training purposes.	**Given** access and downloading the dataset. **When** I explore the dataset and inspect the downloaded files. **Then** I should find a diverse collection of images including blur and bokeh product photographs and the dataset should be well-organized and annotated with labels indicating whether each image is a blur, bokeh, or normal.	High

Title	User Story	Acceptance Criteria	Priority
Train Model	As a user, I want to train a machine learning model that accurately distinguishes between blurry, bokeh, and normal display product images.	**Given** access to the dataset and training resources. **When** I train the model. **Then** it s hould achieve a high accuracy rate in classifying images as blurry, bokeh, and normal display product images.	High
SDK Integrating	As a user, I want to integrate the classification tool into my applications using a Software Development Kit (SDK).	**Given** integrated SDK. **When** I use its methods for image processing and classification. **Then** it should offer simplicity and intuitive interfaces.	High

Title	User Story	Acceptance Criteria	Priority
API For Classifying Images	As a user, I want to utilize a web API for classifying images remotely.	**Given** access to the API documentation and endpoints **When** I interact with the API. **Then** it should offer endpoints for submitting image data and receiving classification results.	High